A few flows carry almost all of the bytes

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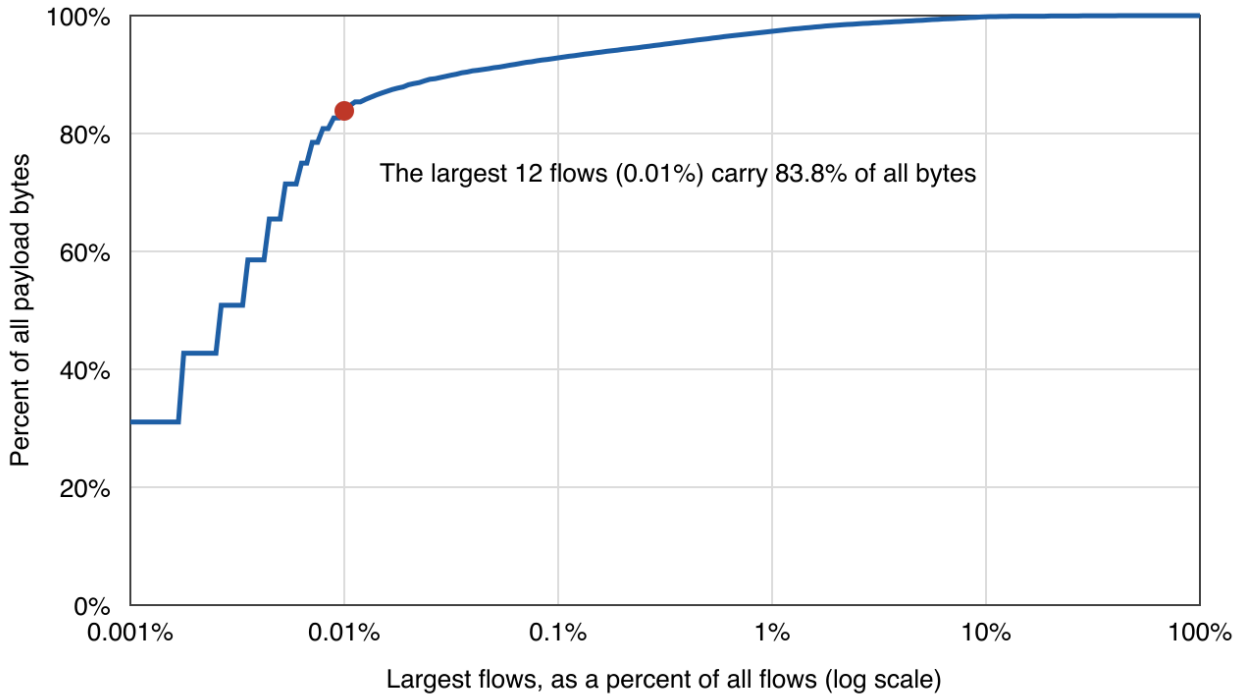


Figure 1. Percent of all payload bytes carried by the largest flows, with flows sorted from largest to smallest.

Payload bytes in this trace are spread very unevenly across flows. I ran my packet trace program in its flow mode on the class trace, and it found 116,009 TCP and UDP flows carrying 1,829,666,291 payload bytes in total. I then passed that flow output to my flow analysis script, a short Python program included in the tarball, which sorts the flows by payload bytes and draws Figure 1. The largest 12 flows, only 0.01% of all flows, carry 83.8% of the bytes, and the largest 1% of flows carry 97.3%. Ten of those 12 flows involve the same local host, 14.1.187.33. At the other end, 47.6% of all flows carry no payload at all. Many of these look like failed or probing connections to SSH. There were 37,516 TCP flows to port 22, about a third of all flows, yet together they carried only 12,783 bytes, and a single remote host (43.255.191.180) started 14,357 of them. So this network looks very different depending on whether you count flows or bytes. Most flows are tiny probes, while a handful of long transfers make up nearly all of the traffic volume.